Momentum Straddle Strategy: Stepwise Logic, Enhancements, and Implementation Sequence

1. Core Strategy Logic (Steps 1–7)

Step 1: Define the universe (top 20 liquid optionable stocks) and evaluation dates (monthly, 3rd Friday).

Step 2: For each stock and evaluation date, select all call and put options expiring the following month.

- Filter calls: delta 0.25–0.75, lowest bid-ask, positive open interest.
- Find the matching put (same strike, expiration, positive open interest).

Step 3: Construct delta-neutral straddle using calculated weights from the call/put deltas; use mid-prices for simulation.

Step 4: At expiration, calculate the return of each leg, then the weighted straddle return relative to entry cost.

Step 5: Generate a momentum signal for each stock/month as the average of the past 11 months' straddle returns (excluding the most recent month).

Step 6 (Key Change):

- **Old:** Sort into quantile buckets (e.g., quintiles) for long/short.
- **New:** Directly pick top 3 and bottom 3 tickers by momentum as long and short candidates, respectively.
- If a top/bottom 3 ticker lacks a valid straddle return for the next month, move to the next available (e.g., 4th ranked) so both portfolios always have 3 tickers.

Step 7: Evaluate strategy performance: monthly long-short return (difference in average returns of long and short baskets), plus metrics (mean, volatility, Sharpe, t-stat, cumulative P&L).

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2. Performance Enhancements: Stop-Loss and Profit-Booking

A. Stop-Loss Implementation

We tested **two separate stop-loss logics**:

1. **Combined Straddle Stop-Loss:**
    - Monitor the total (combined) straddle position.
    - If the combined straddle's value drops by 50% or 75% from entry, close **both legs** immediately.
    - Long Straddle Return Calculation:

$$\text{entry} = w_C \frac{\text{bid}_{\text{call}} + \text{ask}_{\text{call}}}{2} + w_P \frac{\text{bid}_{\text{put}} + \text{ask}_{\text{put}}}{2}$$

$$\text{combined}_t = w_C \cdot \text{mid}_{\text{call}}(t) + w_P \cdot \text{mid}_{\text{put}}(t)$$

$$\text{if combined}_t \leq (1-\alpha) \times \text{entry at } t_s : \quad \begin{cases} \text{exit\_cost} = \text{combined}_{t_s} \\ \text{stop\_loss} = \text{entry} - \text{exit\_cost} \\ R_{\text{stop}} = \dfrac{\text{stop\_loss}}{\text{entry}} \end{cases}$$

- Where α is the stop-loss threshold (e.g., $\alpha = 0.5$ for 50%).

$$\text{call\_payoff} = \max(S_T - K, 0), \quad \text{put\_payoff} = \max(K - S_T, 0)$$

$$\text{payoff} = w_C \cdot \text{call\_payoff} + w_P \cdot \text{put\_payoff}$$

$$\text{hold\_profit} = \text{payoff} - \text{entry}, \qquad R_{\text{hold}} = \frac{\text{hold\_profit}}{\text{entry}}$$

$$R = \begin{cases} R_{\text{stop}}, & \text{if stop-loss triggered} \\ R_{\text{hold}}, & \text{otherwise (expiration)} \end{cases}$$

2. **Individual Leg Stop-Loss:**
    - Monitor each leg (call/put) **separately** after entry.
    - If a leg's value drops by 50% or 75% from entry, close **only that leg** and keep holding the remaining leg until expiration (unless it also hits its stop-loss).
    - Long Straddle Return calculation:

$$R_{\text{straddle,ILSL}} = \frac{w_C \cdot (S_{C,\text{exit}} - C_0) + w_P \cdot (S_{P,\text{exit}} - P_0)}{w_C \cdot C_0 + w_P \cdot P_0}$$

Where:

- $S_{C,\text{exit}}$ =
  - If call leg hit stop-loss: price when stop-loss triggered
  - Else: payoff at expiration = $\max(S_T - K, 0)$
- $S_{P,\text{exit}}$ =
  - If put leg hit stop-loss: price when stop-loss triggered
  - Else: payoff at expiration = $\max(K - S_T, 0)$

B. Profit-Booking Implementation

- We implemented profit-booking (taking profit and closing positions early if a target return is reached) **separately** from stop-loss.
- Profit-booking was applied at three threshold levels:
  - **50%, 80%, and 100%** for long straddles;
  - **50%, 80%, and 99%** for short straddles.
- For each level, if the combined straddle return reaches the threshold before expiration (regardless of individual leg status), the entire position is closed to lock in gains.

C. Combined Strategies

- After testing individual stop-loss and profit-booking logics separately, we combined the **individual leg stop-loss at 75%** with profit-booking on the combined straddle (at various levels) to further improve risk-adjusted performance.
- This final approach allows for:
  - Early exit of losing legs (risk control),
  - Early profit capture for strong performers (return enhancement),
  - And continued upside participation when appropriate.